HARD STOP 3: Early Implication Validation Package

CISE 699 – Thesis /Project computer info science

Fall - 2026

Instructor: Majid Shaalan, Ph.D.

Submitted by:

Gaurav Sharma

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Harrisburg University of Technology

A quick note on this package before it starts, the same note Hard Stop 2 opened with, because it still applies. This assignment (RU-06, Hard Stop 3) is written mostly for software, machine-learning, agentic-AI, and cybersecurity-build projects, where "the data pipeline, toolchain, or core logic" means running code against real data, and "early technical evidence" means logs and test output from that code. My project doesn't build anything, so every section below is a direct translation: the "baseline workflow" is my literature search-and-screening protocol, "data ingestion" is running the finalized search string against a live database, the "toolchain" is Zotero, Litmaps, and my seven database sources, and "API integrations" is the Zotero Connector capturing citation metadata. I say this once here so I don't have to repeat it in every section.

Confirmed dates: Hard Stop 1 (Proposal Approval) was submitted September 13, 2026. Hard Stop 2 (Design Review Package) was submitted September 27, 2026. This package builds directly on the Implementation Baseline and Engineering Readiness Checkpoint work already completed and tagged as release v1.0-baseline in my repository on October 4, 2026, corrected and extended with the evidence in this package as of today, October 11, 2026.

1. **AI Usage Log**

Entry #1: Took path to Fixing the Evidence File (using Claude).

This tool was used to fix the Evidence File's smoke-search.png, which was originally a screenshot of my Zotero window, not the correct Google Scholar results page. It re-ran the search, created correct screenshots and added it to the GitHub repository for the Evidence File. It also used the same method to delete four unused placeholder.txt files.

Entry 2 — Building the Screening Tracker

AI Tool: Claude (Anthropic AI assistant), Python

Log full-scale source screening in a spreadsheet of the same format as used in the five-source pilot tracker.

Entry 3 — Mapping This Assignment to My Project

AI Tool: Claude (Anthropic AI assistant)

Identified data pipeline, toolchain, AP and mapped to research project / Thesis terms.

1. **Toolchain and Data Validation**

For RU-06 Criterion 1, since we have no data pipeline, API or runtime for these review items, the ones below list the review protocol's tools, and then we have the literature for a dataset instead.

2.1 Machine and Environment

All tools are being run from the same environment: a personal computer running Zotero desktop 7.x software, Lit maps and Microsoft Word. Institutional single sign-on is available for the university's licensed databases. A GitHub repository has also been set up for the artifacts generated by this checkpoint (gsharma1989/CISE--699). No GPU or server is available for runtime testing of any kind, and this has therefore been re-checked rather than simply re-stated.

2.2 Data Ingestion- the search protocol, Run Live

Using the search string "'artificial intelligence' AND 'cybersecurity' AND 'risk management'" I ran the search for Google Scholar twice today. I ran the search for two different browser sessions. The first was to create the corrected evidence file for the rubric's stated common failure mode of a one-off manual search that cannot be reproduced. I then immediately ran the search from a completely fresh tab with no shared state between the two. Surprisingly both returned 80,700 results with the top 5 results being in the exact same order (NIST AI RMF, Beriša et al., Islam et al., Ouariach et al., and related work). This search was intentionally re-run to confirm that it could be reproduced and thus treated as proper evidence.

2.3 Toolchain Assumptions validated

Assumption	**Validation Result**
Zotero Connector correctly parses a live Google Scholar result into structured metadata.	Confirmed in Smoke Test 3 (Section 3.3): title, full author list, and year captured correctly with nothing typed by hand.
GitHub web upload correctly replaces a file in place and preserves commit history.	Confirmed today: evidence/smoke-search.png was replaced via a new commit (not a new filename), and the prior incorrect version remains visible in file history for audit purposes.
Screening-tracker spreadsheet formulas recalculate correctly without manual totals.	Confirmed: 135 formulas across the tracker recalculated with LibreOffice's formula engine, zero errors reported.
Institutional single sign-on still provides working access to licensed databases.	Not re-tested this round; last confirmed in Hard Stop 2, Section 3.5. Flagged in Section 6 below as not yet re-verified.

2.4 Architecture Mapped to the Design

The rubric's tools table asks for a listing of the "architecture" for the eight-stage conceptual framework for the intervention laid out in Hard Stop 2, Section 2.2. Only the front end of the chain has been touched technically so far: AI Adoption (just as the framing trigger for the tools is not testable) and Threat Identified (the intervention indirectly uses pilot search and pilot screening for literature that has been tagged into the seven categories for threats and risks for the framework. The remainder of the architecture, Control Selection, Governance & Compliance Obligation, Implementation Evidence, Effectiveness Evidence, and Continuous Monitoring, have not yet been touched by any of the suggested smoke tests. This is discussed further in Section 5.

2. Early Technical Evidence and Correctness

In RU-06 Criterion 2, the three smoke tests of the Implementation Baseline checkpoint (3.1–3.3) of the Requirements to be implemented by Search (3) are re-stated here complete with evidence for the 3 tests, including the correction noted today.

3.1 Smoke Test 1 — Pilot Search

To complete the work for Smoke Test 1 — Pilot Search, I completed a test search using the completed search string in Section 2.2 to test the search against Google Scholar to locate the NIST AI RMF document. Evidence for this test can be found in the file evidence/smoke-search.png in the repository for this report. After re-running the search string as part of my correction for Entry 1, the evidence for this test is now a correct screenshot of the results page from Google Scholar with a query box to search for AI papers and “About 80,700 results” and the NIST document at the top of the list. Result: pass.

3.2. Smoke Test 2 — Pilot Screening

The first five results of the search string have been screened against the inclusion/exclusion rule (on topic for AI cybersecurity risk management and not primary data collection, low-level algorithmic detail or operationally actionable attack detail, within the scope boundaries of the project charter). Evidence: screening/screening-pilot.xlsx with five rows, each result having a title, year, decision and reason for inclusion for each result within the project’s framework stages. All five sources included. Note however that five results do not constitute evidence of the full range of 40-80 potential sources, only that the screening process has been correctly applied to the results of the search string.

3.3 Smoke Test 3 — Citation Tool (“API” Equivalent)

This smoke test involved saving a single citation from a Google Scholar results page to a Zotero library using the Zotero Browser Connector. The resulting saved item is shown in evidence/smoke-connector.png. All fields (title, authors, year, etc.) were filled in correctly. Result: pass.

3.4 Evidence Summary

Test	Evidence File	Result	Repeated?
Pilot Search	evidence/smoke-search.png	Pass	Yes, twice today, consistent result count and top results
Pilot Screening	screening/screening-pilot.xlsx	Pass	Not yet repeated at full scale (Section 5)
Citation Tool	evidence/smoke-connector.png	Pass	Not re-run today; last confirmed Oct 4, 2026

3. Experimental Discipline and Reproductivity

RU-06 – Criteria for artifacts, experiment tracking, configuration, provenance and reproducibility.

4.1 Version Control as Configuration Control

I store all of the artifacts for this project within a single GitHub repository (github.com/gsharma1989/CISE--699) at the release for the baseline version (v1.0-baseline, e09a3c7, Oct 4, 2026). There have been 18 changes on top of this tagged baseline (through today), each with a very descriptive commit message (not just ‘update’), including the four today: “Replace smoke-search.png with correct Google Scholar screenshot” and “Delete placeholder.txt” for the four folders. Since a build project’s git history is supposed to demonstrate exactly what was done and why, I have done the same for this project.

4.2 Data Provenance

Every screened reference, whether pilot or full-scale, is recorded with the database from which the reference was retrieved (the new Database/Source column in the full screening tracker, Section 4.3 below) not simply the reference’s title and screening decision.

4.3 Experiment Tracking — the Full-Scale Screening Tracker

The updated tracker has been built as a 4-tab workbook (Instructions, Screening Tracker, Progress Summary, Spot-Check Log) – the Screening Tracker has a new Database/Source column and has data-validation on the Category and Decision dropdowns. The Progress Summary now has formula-driven progress counts (total screened, included and excluded as well as by category) so one can easily track progress towards the 40-80 paper target. All 135 formulas re-calculated perfectly without any errors. The workbook has been generated and is ready to be committed to the /screening/ directory. This has been listed as an action item in Section 6.

4.4 Clean-Restart Reproducibility Self-Check

The self-checking rubric for hard stop 3 was answered for the pilot search results in positive fashion today (i.e. same results count and even the top results) for two separate, independent runs using two different browsers for each. The answer for the full environment was established in Hard Stop 2 (Section 3.5) and still holds: all database sources (same 7 as used for the rest of the program) are all web-based (Lit maps, etc.) or free cross-platform software (Zotero), so complete reproduction from scratch on another day/another machine by another person is possible and was confirmed today.

4. Interpretation of Early Results

This section covers RU-06 Criterion 4. The emphasis in this assignment is to be as honest as possible and not try to polish up things that are not working rather explain the evidence in all its glory.

5.1 What Evidence Actually Shows

To demonstrate that the review protocol is technically alive at its front end I wanted to show that the search string returns a large, relevant and reproducible set of results on a freely accessible live database. The screening workflow has been applied to five ‘real’ results in order to test that it produces a clear decision and documented reason for each of them. The citation-capture tool is used to capture citation information from live results on the database, without the need for any manual typing and this information is populated into the relevant, structured metadata for the results.

5.2 What the Evidence Does Not Yet Show

The fact that we have gone through five sources of information does not necessarily mean that the rest of the 40-80 sources will be treated in the same manner. Full-scale search and screening of all 7 databases have not yet commenced. Of greater concern is the fact that none of the smoke tests have yet to test the framework-validation aspect of the work. This aspect involves a simple mapping of a control to a particular clause in the governance of a compliance obligation. This work is all contained within Control Selection and Governance & Compliance Obligation, the two untested sections of Section 2.4. Hard Stop 2, Section 6.2 identified this as the single highest risk dependency in the project and, yet nothing has been done to reduce this risk. In fact, all that has been done is to test the prior work and find it to work as expected.

5.3 An Anomaly, Named Plainly

What the evidence failed to show for Smoke Test 1 was a screenshot of my Zotero window listing the Google Scholar search results instead. That could have been an error on my part while juggling multiple open windows. In any case, the evidence, once corrected, revealed that the search worked and that evidence capture needs to be double-checked as evidence to verify that it was indeed captured correctly. I shall continue to verify all evidence capture from now on.

5.4 Single Technical Risk Most Likely to Derail the Midpoint Demo

The only part of the thesis remaining with high risk as defined in Hard Stop 2, Section 6.2 is the framework-validation mapping of the material in Control Selection to the corresponding governance clause in Governance & Compliance Obligation. There is no technical evidence to date that this mapping has been tested. Therefore, the de-risking plan put in place in Hard Stop 2, Section 6.2 (to pilot the mapping of the material in Control Selection to the appropriate clause in Governance & Compliance Obligation on five to eight sources that have already gone through the screening process for the thesis) remains and will be executed in weeks 9 and 10 of the build order.

5. Defect, Bottleneck, and Workaround Log

This information has been pulled directly from the live risk and known-issues logs found within the repository (logs/risk-log.md and logs/know-issues.md). It has not been re-written for this package and therefore reflects the current situation and not a summary of the situation after the fact.

6.1 Risk and Issue Log

ID	Description	Severity	Status
R-01	Loss of access to a licensed database mid-review.	Medium	Mitigated
R-02	Single-reviewer screening bias, no second coder.	High	Open, pending supervisor sign-off
R-03	Governance references are still in draft or moving.	Medium	Mitigated
R-04	English-language and database-coverage bias.	Low	Accepted, disclosed
R-05	Framework-to-governance mappings may not be one-to-one.	Medium	Mitigated by design
R-06	Pilot search or pilot screening turns up a protocol problem.	Medium	Mitigated; pilot found no issues

6.2 Known Issues

- Single-reviewer screening spot-checks are tentatively fine with only one reviewer but still need confirmation from my supervisor whether that’s okay without a second coder.

- Framework-validation pilot (mapping controls to governance clauses) has not yet started; will start in week 9-10.

- Institutional single sign-on to licensed databases (IEEE Xplore, ACM Digital Library, Scopus, Web of Science) was not re-tested for this checkpoint and thus has not been re-confirmed since Hard Stop 2.

- The full-scale screening tracker has not yet been committed to the repository in Section 4.3.

7. README and Environment Notes

This is still accurate for the contents of the repository for the thesis on AI-driven cybersecurity risk management for CISE 699. Nothing has changed structurally with the evidence files for the thesis (other than for one file which has now been corrected), and the four unused placeholder files have been deleted from /evidence/, /protocol/ and /screening/. The environment notes for this Section 2.1 are accurate for now and have been recorded in the repository's own logs/changelog.md file.

8. Supervisor Meeting and Scope Check

The only real open item is the meeting with the supervisor in Section 8 above, for discussion of whether any scope or sequencing changes are required before the midpoint demo. Specifically, I'd like to get his take on whether or not to move up the framework-validation pilot (currently Hard Stop 2, scheduled for weeks 9–10) since this will be the only stage with no supporting technical evidence and I'd like to get his take on whether or not he thinks it should be moved up and validated sooner rather than later. I'd also like to discuss the still-open sign-off for R-02 in the risk log above, as this will also require discussion with the supervisor.

9. Action Checklist Before Submission

- smoke-search.png fixed evidence in repository /evidence/ (corrected original file, not cached preview) (done).

- All 4 of the placeholder.txt files were removed from /evidence/, /protocol/, and /screening/ today.